

Name: _____

Double Award Science – Physics Unit 1 Higher Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /60	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries- Raw Score

Year	E	D	C	B	A	A*	Max
2017	2	6	13	20	28	36	44
2018	3	5	10	17	35	33	41
2019	6	9	15	21	28	35	42
2022	3	5	11	18	26	34	42
2023	6	10	16	23	34	38	45

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3430UC0-1



SCIENCE (Double Award)

Unit 3: PHYSICS 1
HIGHER TIER

MONDAY, 19 JUNE 2017 – MORNING

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	15	
2.	8	
3.	9	
4.	10	
5.	4	
6.	14	
Total	60	

ADDITIONAL MATERIALS

In addition to this examination paper, you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space use the continuation page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question 6(a) is a quality of extended response (QER) question where your writing skills will be assessed.



JUN173430UC0101

Equations

current = $\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
total resistance in a series circuit	$R = R_1 + R_2$
total resistance in a parallel circuit	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$
energy transferred = power $\times$ time	$E = Pt$
power = voltage $\times$ current	$P = VI$
power = current ² $\times$ resistance	$P = I^2R$
% efficiency = $\frac{\text{energy (or power) usefully transferred}}{\text{total energy (or power) supplied}} \times 100$	
density = $\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
units used (kWh) = power (kW) $\times$ time (h) cost = units used $\times$ cost per unit	
wave speed = wavelength $\times$ frequency	$v = \lambda f$
speed = $\frac{\text{distance}}{\text{time}}$	

SI multipliers

Prefix	Multiplier
p	1×10^{-12}
n	1×10^{-9}
μ	1×10^{-6}
m	1×10^{-3}

Prefix	Multiplier
k	1×10^3
M	1×10^6
G	1×10^9
T	1×10^{12}



Answer all questions.

1. The Energy Saving Trust aims to help everyone save energy every day in order to reduce the environmental damage that energy production does to the environment. It provides advice to households and businesses and gives tips for energy savings such as not leaving appliances on standby and choosing the most energy efficient appliances.

The efficiency of washing machines depends on how they are used as they can wash clothes at a variety of temperatures. Many clothes retailers and washing detergent manufacturers recommend washing clothes at 30 °C in order to be more energy efficient. According to the Energy Saving Trust “*Washing clothes at 30 °C rather than at higher temperatures uses around 40 % less energy.*”

The table below gives data for different washing temperatures for a particular washing machine.

Washing temperature (°C)	Mean power (W)	Wash time (minutes)
30	500	60
40	500	90
50	500	120

- (a) (i) Use equations from page 2 to calculate the cost of washing clothes at 40 °C. The cost of one unit of electricity is 16 p. [4]

Cost = p

- (ii) The Energy Saving Trust suggests that washing clothes at 30 °C reduces energy use by at least 40 % compared to washing at higher temperatures. Use the data in the table to investigate the validity of this claim. [3]

Space for calculations

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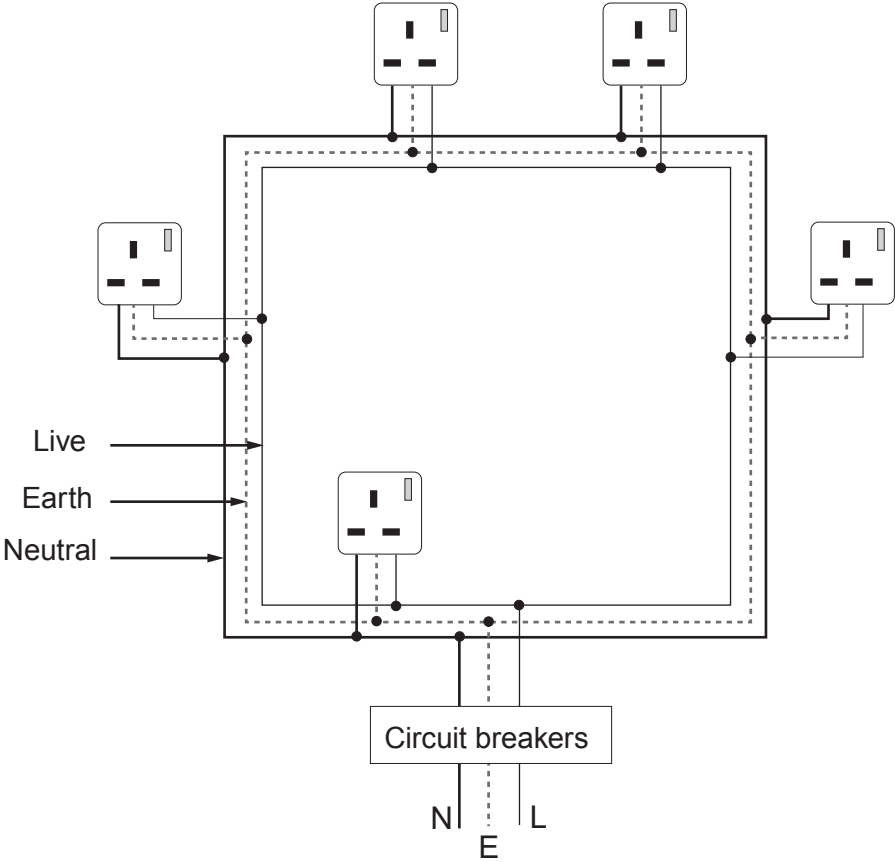
(iii) Washing clothes at 30 °C saves money on household bills. Explain why reducing energy consumption also helps the environment. [2]

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(b) The washing machine is connected to the ring main shown below. The voltage in the ring main is 230 V.



(i) State which wire the fuse must be connected to. [1]

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(ii) Describe the function of the earth wire. [2]

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Examiner
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- (c) When the voltage is 230V, the mean power of the washing machine during a wash is 500W. However when heating the water it must be able to produce a maximum power of 2400W. The householder has 3A, 5A and 13A fuses available and chooses to fit a 3A fuse into the plug. Investigate whether or not this is a suitable choice. [3]

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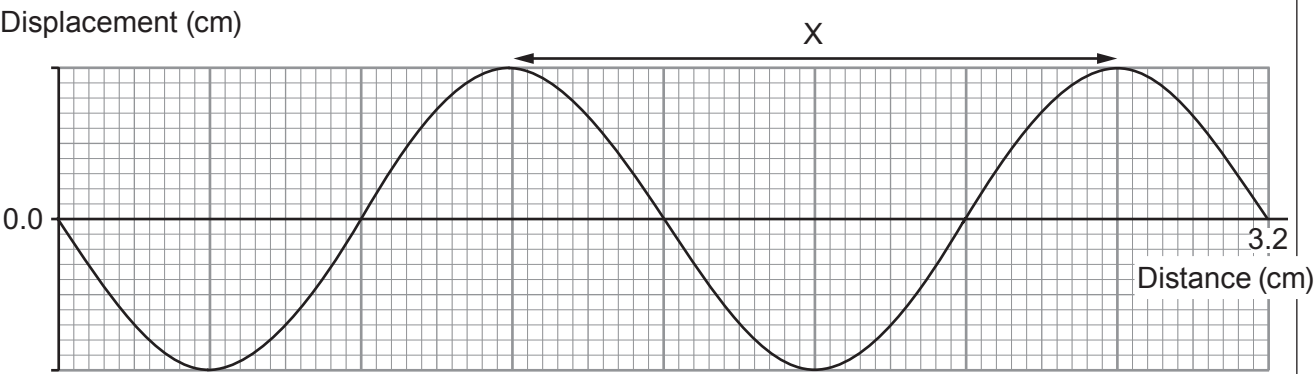
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2. A teacher demonstrates water waves in a ripple tank.

(a) The diagram represents the waves produced at a particular frequency.



(i) Use the information in the diagram to calculate the distance X. [2]

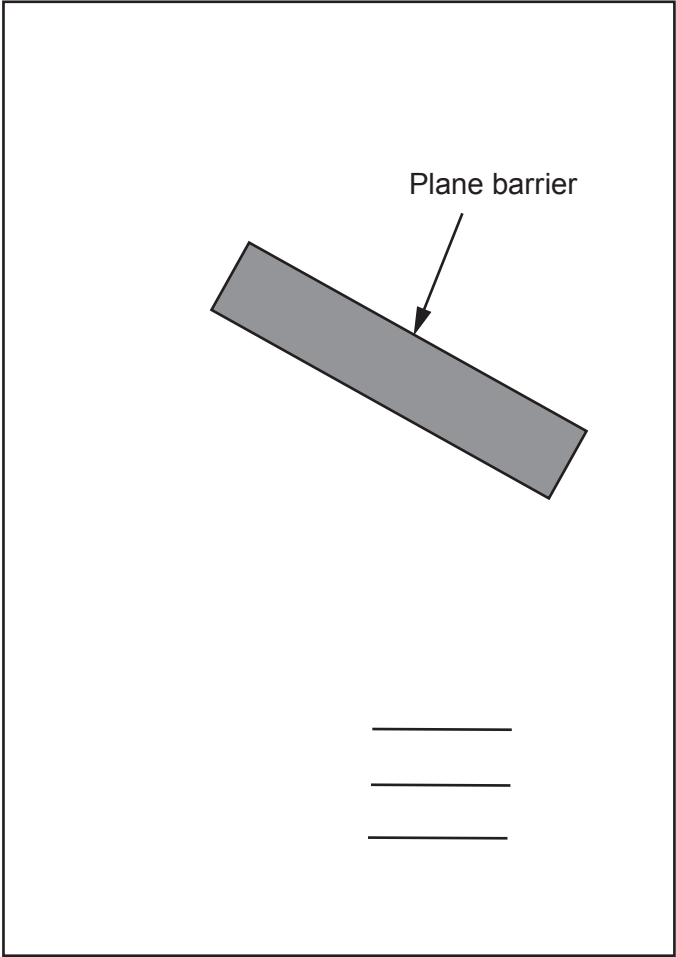
X = cm

(ii) A wave takes 5 s to travel 40 cm along the surface of the water in the ripple tank. Use equations from page 2 and the information above to calculate the frequency of the wave. [4]

Frequency = Hz



(b) Wavefronts travel towards a plane barrier. **Complete the diagram** to show the path of the wavefronts as they are incident on and reflect off the barrier. [2]



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3. (a) A student is investigating the current-voltage characteristic of a filament lamp. In the space below, **draw a circuit diagram** of the circuit she would use. [3]

(b) The student collects the following results.

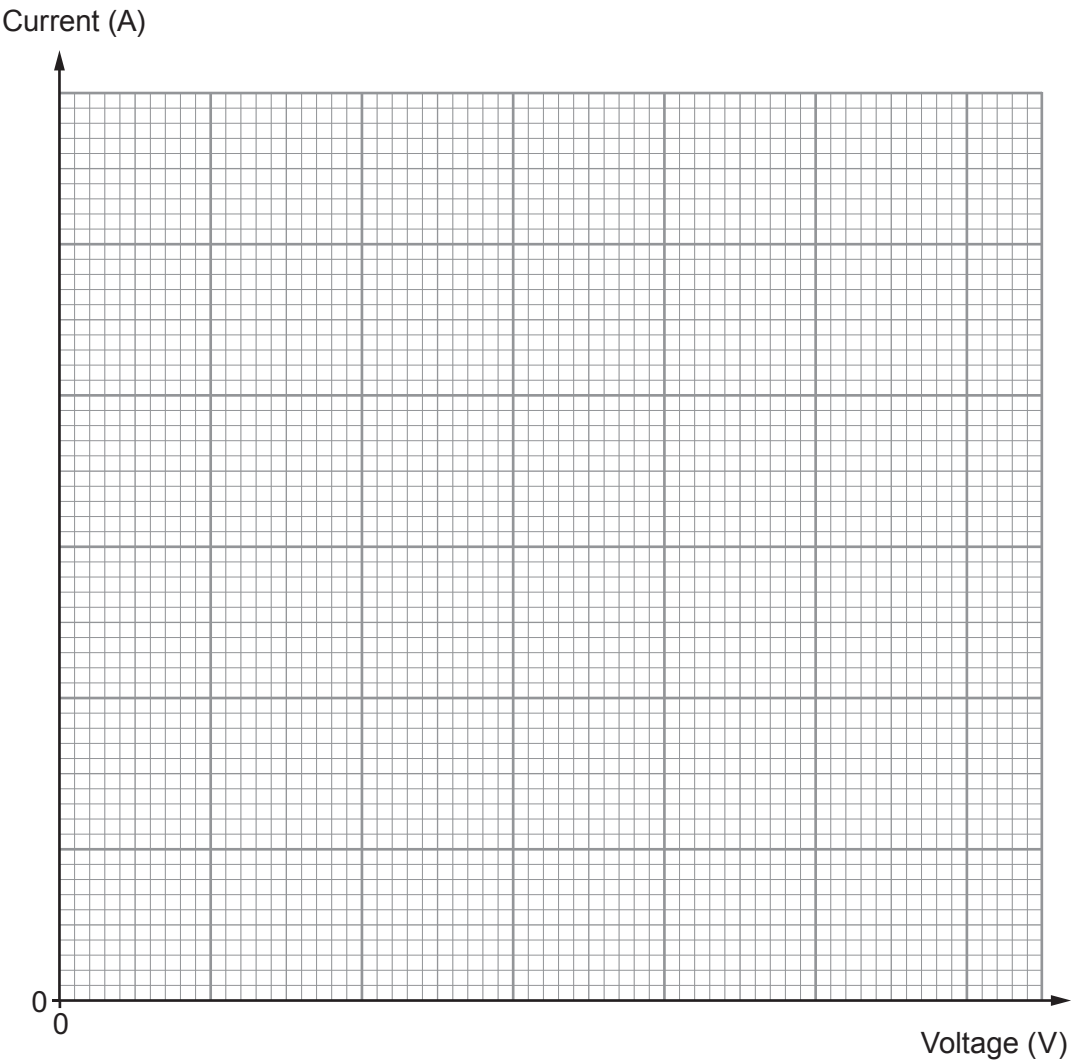
Voltage (V)	Current (A)
0.0	0.00
2.0	0.40
4.0	0.76
6.0	0.94
10.0	1.08
12.0	1.10



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(i) Plot the data on the grid below and draw a suitable line.

[3]



(ii) She concludes that the resistance of the lamp decreases as voltage increases. Explain using calculations, whether you agree or disagree with this conclusion. [3]

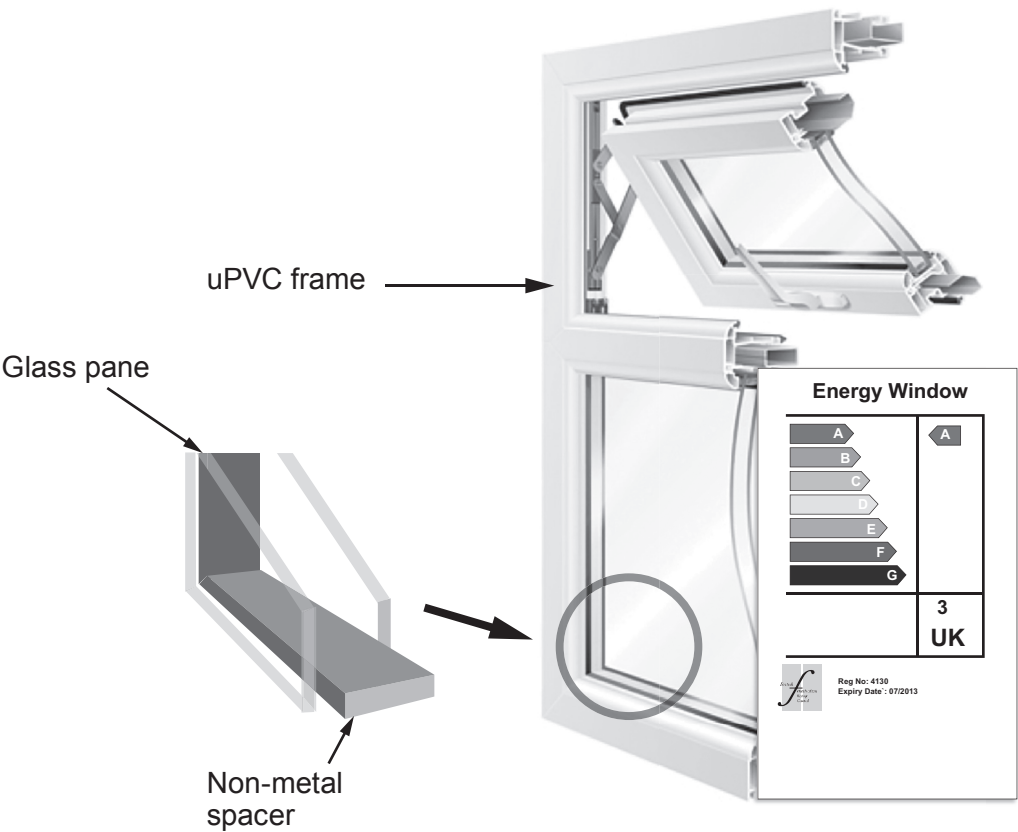
Space for calculations

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4. All properties lose heat through their windows and roof. Energy-efficient installations such as uPVC windows and loft insulation keep homes warmer as well as reducing energy bills.
- (a) A homeowner replaces aluminium framed double glazed windows which contain metal spacers between the panes of glass with uPVC windows. This new frame is made of uPVC and the window panes are separated by non-metal spacers.



- (i) Explain why the aluminium window loses heat at a faster rate than the new uPVC window. Include reference to particles in your answer. [4]

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Examiner
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(ii) The photograph shows an uninsulated loft.



Explain how heat is lost through the roof of a house when the loft is not insulated.
You should refer to the ceiling, the loft space and the roof tiles in your answer. [3]

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(b) Explain why reducing energy losses from our homes benefits the environment. [3]

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5. Electromagnetic (em) waves are used extensively in a wide variety of applications. The table below gives information about different regions of the electromagnetic (em) spectrum. It shows ranges for wavelengths and frequencies.

Region	Wavelength (m)	Frequency (Hz)
radio	$> 1 \times 10^{-1}$	
microwave	1×10^{-3} to 1×10^{-1}	3×10^9 to 3×10^{11}
infra-red		3×10^{11} to 4×10^{14}
visible		4×10^{14} to 7.5×10^{14}
ultraviolet	1×10^{-8} to 4×10^{-7}	7.5×10^{14} to 3×10^{16}
X-ray	1×10^{-11} to 1×10^{-8}	3×10^{16} to 3×10^{19}
gamma ray	$< 1 \times 10^{-11}$	

(a) Complete the table above using the ranges below. [3]

$(7.5 \times 10^{-7}$ to $1 \times 10^{-3})$ $(< 3 \times 10^9)$ $(> 3 \times 10^{19})$ $(4 \times 10^{-7}$ to $7.5 \times 10^{-7})$

(b) All em waves transfer energy. The shortest waves are the most ionising. The table below shows energy values in an **incorrect order**. Arrange the energy values in the **correct order**. [1]

Region	Energy (J) in incorrect order	Energy (J) in correct order
radio	$> 2 \times 10^{-14}$	
microwave	3×10^{-19} to 5×10^{-19}	
visible	2×10^{-17} to 2×10^{-14}	
X-ray	2×10^{-24} to 2×10^{-22}	
gamma ray	$< 2 \times 10^{-24}$	



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6. Consumers rely on a continuous supply of electricity from the National Grid system.

(a) Describe how the National Grid provides a reliable supply of electricity as efficiently as possible. [6 QER]

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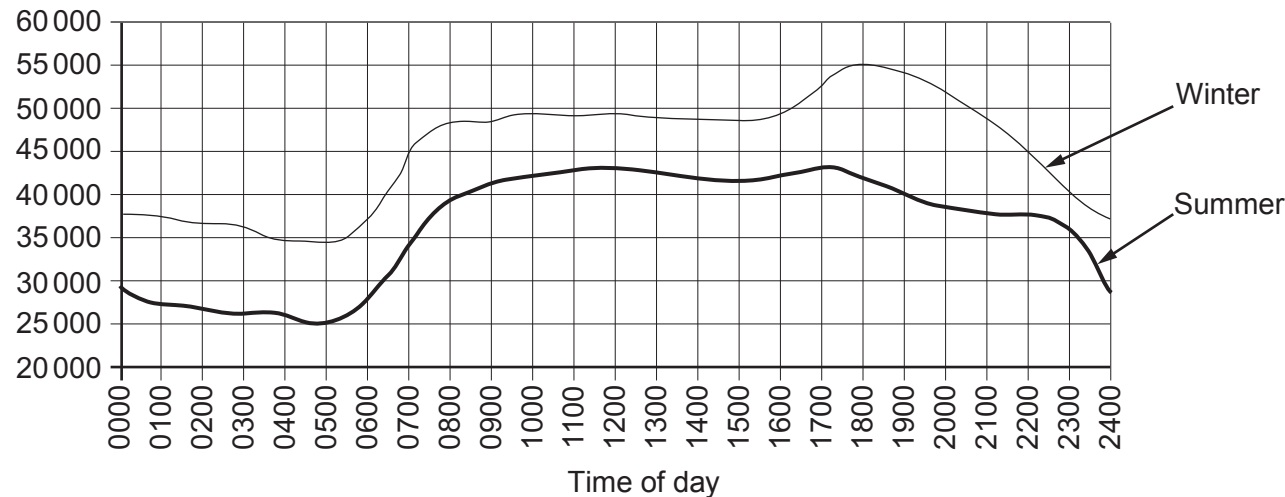
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(b) Information about the National Grid is shown below.

Typical summer and winter demands

National Grid demand (MW)



Network size

- Maximum demand 63 GW
- Output capacity 79.9 GW
- Annual mean electrical energy used in the UK 360 TWh
- Length of 400 kV grid 11 500 km
- Length of 275 kV grid 9800 km
- Length of 132 kV (or lower) grid 5 250 km

Energy losses at maximum demand

- Heating in cables 860 MW
- Fixed losses 260 MW
- Sub-station transformer heating losses 145 MW
- Generator transformer heating losses 155 MW

Total losses = 1 420 MW

(i) Use the information above and an equation from page 2 to calculate the % efficiency of the National Grid system at maximum demand. [3]

% efficiency =



(ii) Use the equation:

power = $\frac{\text{energy transferred}}{\text{time}}$

to calculate the annual mean power used in GW.
(1 year = 8 766 hours)

[3]

Mean power = GW

(iii) Tick (✓) the boxes next to the **two** correct statements about the data opposite. [2]

- At maximum demand there is spare capacity of 16.9 GW ☐
- Maximum demand is equivalent to 85% of total capacity ☐
- The difference in minimum demand between summer and winter is approximately 10 000 MW ☐
- The times of peak demand in summer and winter are **exactly** the same ☐
- The difference in maximum demand between summer and winter is 15 000 MW ☐

END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		0



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SCIENCE (Double Award)

Unit 3 – PHYSICS 1
HIGHER TIER

FRIDAY, 15 JUNE 2018 – MORNING

1 hour 15 minutes

For Examiner’s use only		
Question	Maximum Mark	Mark Awarded
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2.	11	
3.	6	
4.	9	
5.	8	
6.	11	
Total	60	

ADDITIONAL MATERIALS

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INFORMATION FOR CANDIDATES

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The assessment of the quality of extended response (QER) will take place in question **5(b)**.



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Equations

current = $\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
total resistance in a series circuit	$R = R_1 + R_2$
total resistance in a parallel circuit	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$
energy transferred = power $\times$ time	$E = Pt$
power = voltage $\times$ current	$P = VI$
power = current ² $\times$ resistance	$P = I^2R$
% efficiency = $\frac{\text{energy (or power) usefully transferred}}{\text{total energy (or power) supplied}} \times 100$	
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units used (kWh) = power (kW) $\times$ time (h) cost = units used $\times$ cost per unit	
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speed = $\frac{\text{distance}}{\text{time}}$	

SI multipliers

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n	1×10^{-9}
μ	1×10^{-6}
m	1×10^{-3}

Prefix	Multiplier
k	1×10^3
M	1×10^6
G	1×10^9
T	1×10^{12}



Answer all questions.

1. Water waves on the sea where the water is shallower than $\frac{1}{20}$ of their wavelength are known as shallow water waves. The speed of shallow water waves is described by the equation:

$$v = 3.13\sqrt{d}$$

where v is the wave speed (in m/s) and d is the depth of the water (in m).
This equation applies to sea waves whose wavelengths range between 10 m and 150 m.

In regions of the sea where the depth is small, for example near the shore, the speed noticeably changes but the frequency of the waves remains constant.

A shallow water wave is an example of a transverse wave.

- (a) Describe what is meant by a transverse wave. [2]

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- (b) (i) Use the equation above to **complete the table below**. [2]
Space for workings.

Depth of water, d (m)	$\sqrt{d}$	Wave speed, v (m/s)
0	0	0
0.5	0.71	2.21
1.0	1.00	3.13
1.5		3.83
2.5	1.58	4.95
3.0	1.73	5.42
3.5	1.87	
4.0	2.00	6.26



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(ii) Chris suggests that if the depth of the water increases four times, the wave speed doubles. **Use data in the table opposite** to explain whether or not this statement is true. [2]

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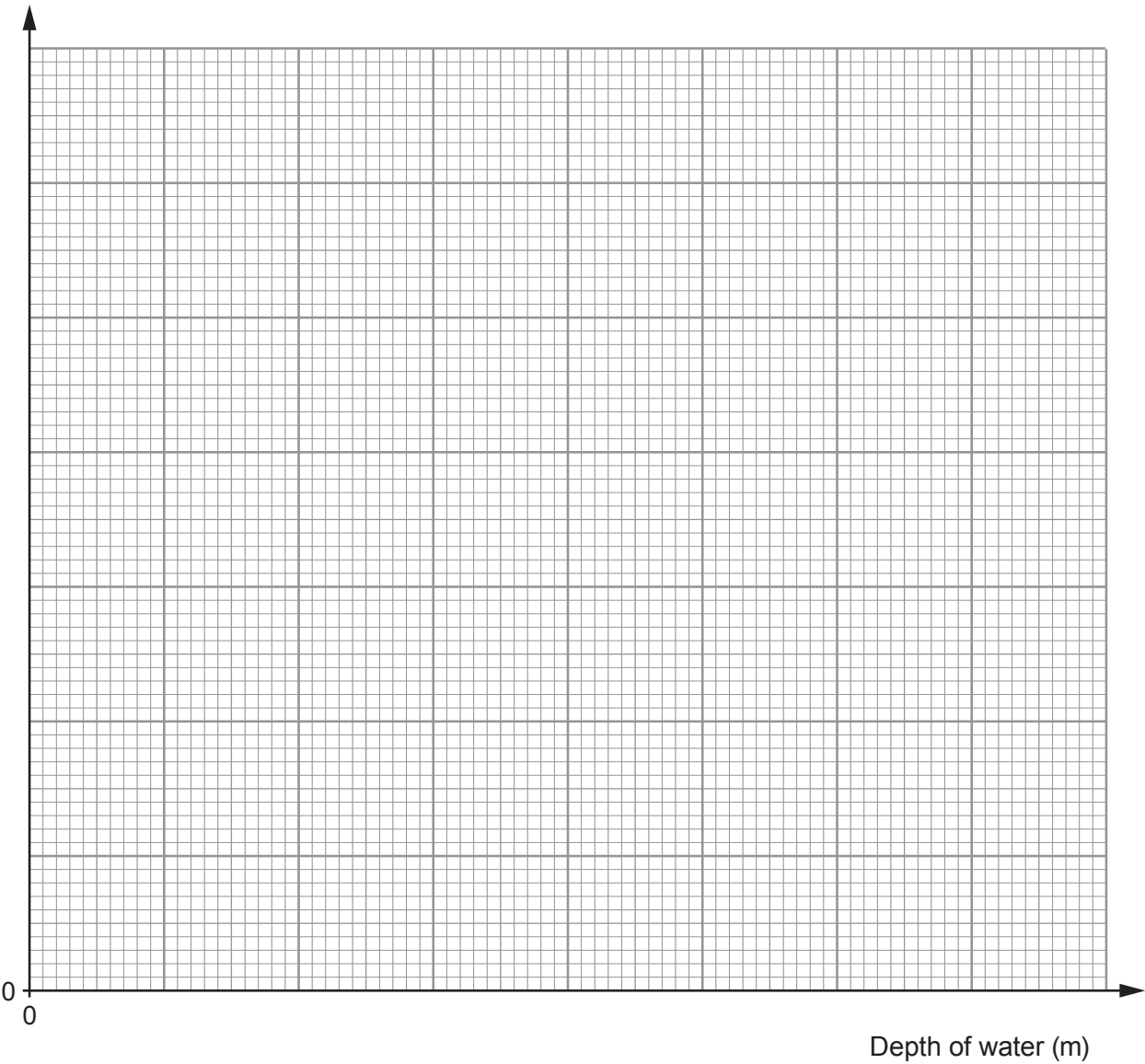
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(iii) Plot the data on the grid below and draw a suitable line. [4]

Wave speed (m/s)



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(c) (i) Use the graph and the equation:

wavelength = $\frac{\text{wave speed}}{\text{frequency}}$

to calculate the wavelength of water waves that have a frequency of 0.2 Hz in water that is 2.0 m deep. [3]

Wavelength = m

(ii) Chris now suggests that as the depth increases, the wavelength decreases. Explain whether this statement is correct. [2]

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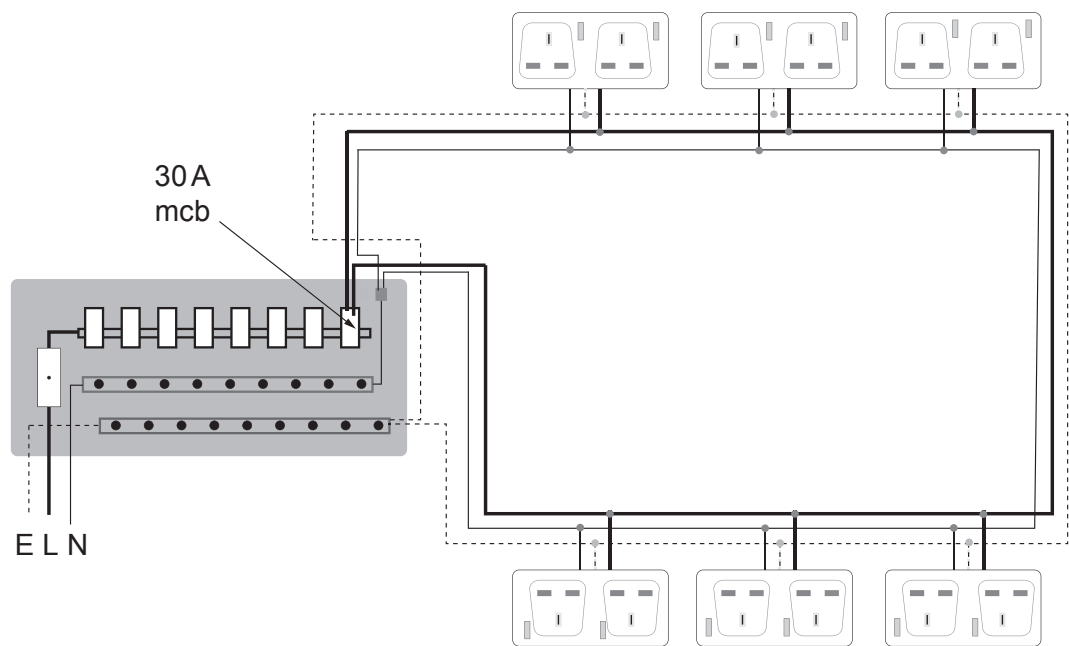
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2. The diagram shows a common domestic ring main circuit.



(a) Give **two** advantages of having a ring main circuit in a house. [2]

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(b) (i) State the function of the live wire. [1]

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(ii) Describe the function of the earth wire. [2]

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(c) The following information is printed on the bottom of a domestic kettle:

TYPE: KBZ3001.BQ
230 V ~ a.c.
2760 W
CAPACITY 1.5L
SERIAL NUMBER 53023

(i) State the differences between alternating current (a.c.) and direct current (d.c.). [2]

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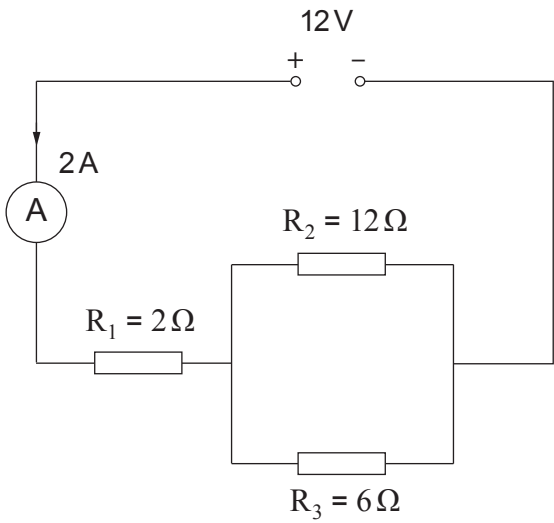
(ii) The voltage of the mains electricity supply in the U.K. is 230V. Use equations from page 2 and information on the kettle to calculate the resistance of the kettle heater. [4]

Resistance = Ω

11



3. A group of students set up the following circuit. Their aim is to measure the current through the ammeter and to use it to calculate the currents and voltages in the various parts of the circuit.



Use an equation(s) from page 2 to answer the following questions.

- (i) Calculate the voltage across R_1 . [2]

Voltage = V

- (ii) Use your answer to part (i) to calculate the current through each of the parallel resistors. [4]

Current through R_2 = A

Current through R_3 = A



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4. In 2015, a householder paid £780 for electricity and £950 for gas. On 1 January 2016 solar panels were installed on the roof to provide **hot water** for the house. The cost of the solar panels was £4 680. In 2016, the householder paid £830 for electricity and £840 for gas.

(a) (i) Gas was used to heat the water in the house in 2015. State how the figures allow you to come to this conclusion. [1]

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(ii) Calculate the expected payback time to recover the cost of the solar panels. [2]

Payback time = years

(iii) Explain how an increase in the cost of a unit of electricity **and** a unit of gas would affect the payback time. [3]

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(b) During the two years, the cost of a unit of electricity did not change but the number of **extra units of electricity** used in 2016 was 300 kWh. Use an equation from page 2 to calculate the cost of a unit of electricity (in pence) during 2015-2016. [3]

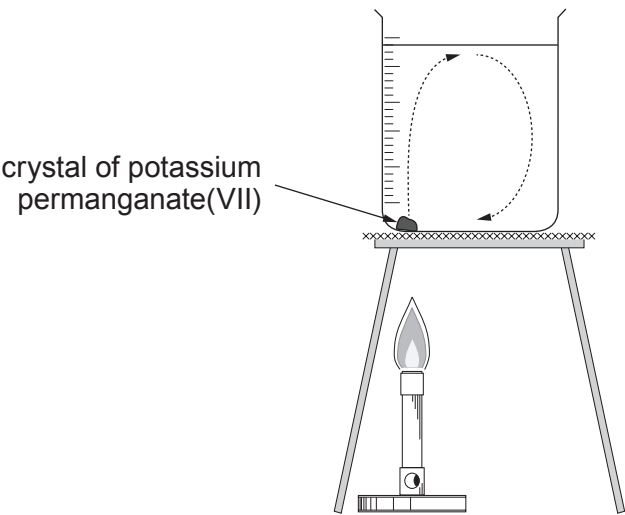
Cost of unit = p

9



5. A class carries out experiments to study heat transfer in solids and liquids.

- (a) In one experiment, the process of convection is shown by colouring some water with potassium permanganate(VII) crystals in the bottom corner of a beaker. The water is then heated and a clockwise convection current is set up.



Explain why the coloured water moves in the way described above.

[2]

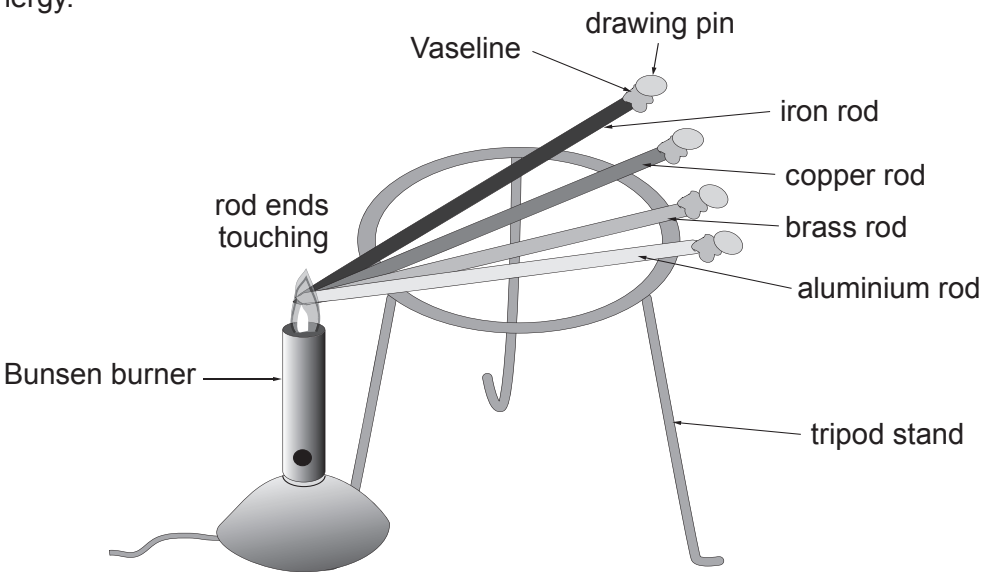
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- (b) In another experiment, four metal rods are tested to compare their ability to conduct heat energy.



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The outcome of the experiment is as follows:

- copper (best conductor)
- aluminium
- brass
- iron (poorest conductor).

Explain in detail, the process of conduction in **copper** in terms of the motion of **particles**.
[6 QER]

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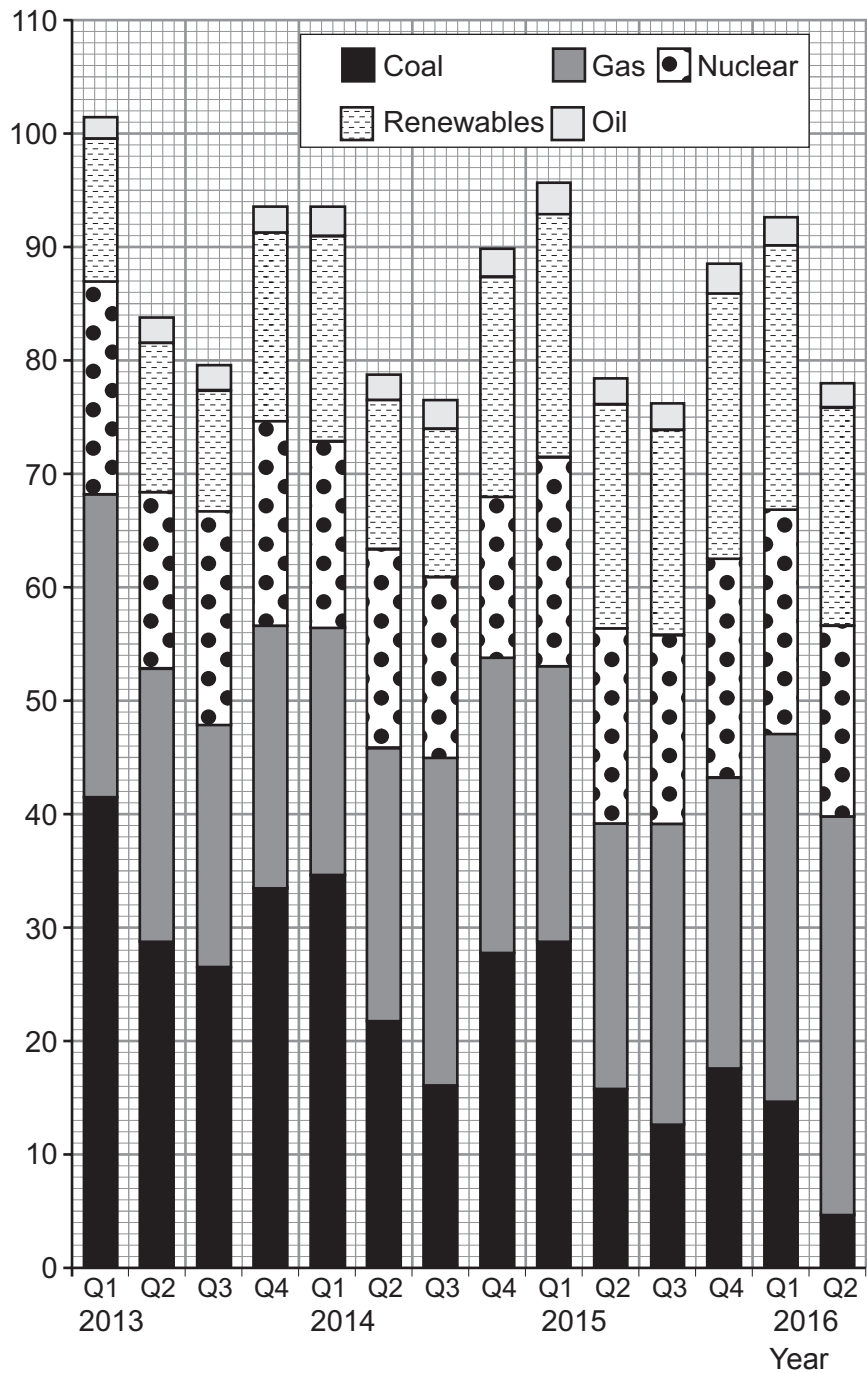
8



6. The Climate Change Act established a target for the UK to reduce its 1990 CO₂ emissions by at least 80 % by 2050. This target represents an appropriate contribution from the UK towards the agreed reduction in CO₂ emissions globally. It aims to limit the global temperature rise to as little as possible above 2 °C.
- To ensure that regular progress is made towards this long-term target, the Act also established a system of five-yearly carbon budgets, to serve as stepping stones on the way.

The diagram below shows the sources of fuel used in the production of electrical energy between the first quarter of 2013 and the middle of 2016.

Generation (TWh)



KEY:

Q1 is 1st quarter of the year (Jan – Mar)

Q2 is 2nd quarter of the year (Apr – June)

Q3 is 3rd quarter of the year (July – Sept)

Q4 is 4th quarter of the year (Oct – Dec)



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- (a) (i) Use the information in the diagram to describe trends in the use of individual fossil fuels and the total use of fossil fuels over the time shown. [3]

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- (ii) The use of nuclear fuel for the production of electricity between Q1 2013 and Q1 2016 remained roughly constant. Suggest possible reasons for this. [2]

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- (iii) Production of electricity from renewable sources remained roughly constant for Q2 2015 and Q2 2016. However, there was a reduction in the contribution by wind and a change in the contribution from solar power. Describe how the weather conditions were different in Q2 2016 compared to Q2 2015. [2]

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- (b) The first four carbon budgets, leading to 2027, have been set in law. The UK is currently in the third carbon budget period (2018-22). Meeting the fourth carbon budget (2023-27) will require that emissions be reduced by 50 % on 1990 levels by 2027.

The 1990 levels of CO₂ emissions were 3900 million tonnes.

Budget	CO ₂ emission targets from all sources	% reduction below 1990 levels for all sources
1st 5 yr period achievement (2008-12)	Down to 3018 million tonnes of CO ₂ by the end of 2012	
2nd 5 yr period target (2013-17)	Down to 2782 million tonnes of CO ₂ by the end of 2017	
3rd carbon target (2018-22)	Down to 2544 million tonnes of CO ₂ by the end of 2022	
4th carbon target (2023-27)	Down to 1950 million tonnes of CO ₂ by the end of 2027	50 % by the end of 2027

- (i) Calculate the targeted percentage drop in CO₂ emissions **from all sources** between 1990 and the end of 2022. Show your workings clearly. [2]

Percentage drop =

- (ii) Between 1990 and the end of 2022, the mean percentage drop in the use of **fossil fuels in the production of electricity** is expected to be 40 %. Comment on the progress being made in achieving the government’s overall target for CO₂ emissions by 2027. [2]

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END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		0



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SCIENCE (Double Award)



Unit 3 – PHYSICS 1
HIGHER TIER

FRIDAY, 14 JUNE 2019 – MORNING

1 hour 15 minutes

For Examiner’s use only		
Question	Maximum Mark	Mark Awarded
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3.	6	
4.	14	
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6.	8	
Total	60	

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Equations

current = $\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
total resistance in a series circuit	$R = R_1 + R_2$
total resistance in a parallel circuit	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$
energy transferred = power $\times$ time	$E = Pt$
power = voltage $\times$ current	$P = VI$
power = current ² $\times$ resistance	$P = I^2R$
% efficiency = $\frac{\text{energy (or power) usefully transferred}}{\text{total energy (or power) supplied}} \times 100$	
density = $\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
units used (kWh) = power (kW) $\times$ time (h) cost = units used $\times$ cost per unit	
wave speed = wavelength $\times$ frequency	$v = \lambda f$
speed = $\frac{\text{distance}}{\text{time}}$	

SI multipliers

Prefix	Multiplier
p	1×10^{-12}
n	1×10^{-9}
μ	1×10^{-6}
m	1×10^{-3}

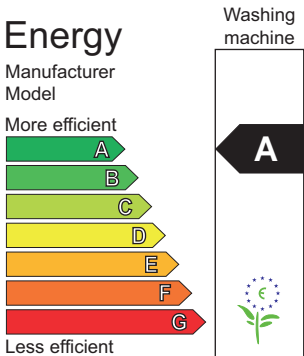
Prefix	Multiplier
k	1×10^3
M	1×10^6
G	1×10^9
T	1×10^{12}



Answer all questions.

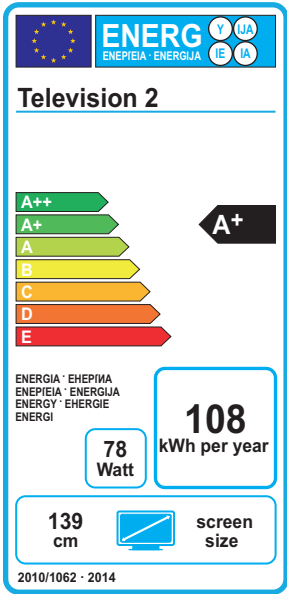
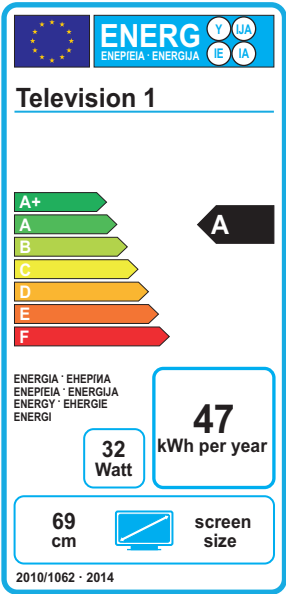
1. Energy rating labels are compulsory on most household appliances such as fridges, dishwashers, washing machines and televisions. These labels allow customers to compare appliances. In addition, the labels give other information about the appliances such as how noisy they are.

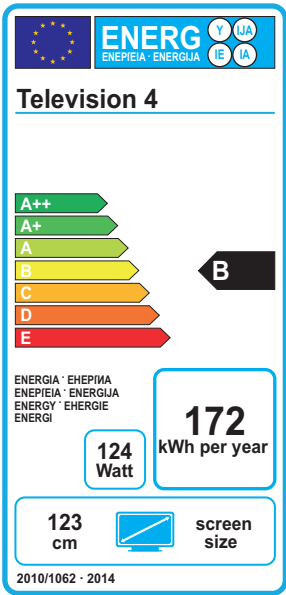
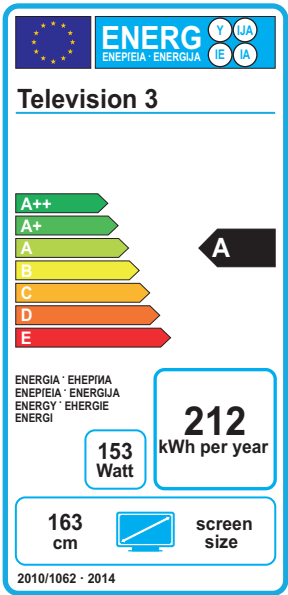
Labels used to rate appliances are from A, the most efficient, to G, the least efficient.



As manufacturers have designed more efficient devices, new categories, e.g. A+, A++ and A+++ have been added to the labels.

The energy rating labels of four televisions are given below and opposite.





Information on the cost of each television is given below.

Television	Purchase cost (£)
1	280
2	1 000
3	1 500
4	800

(a) Use the information from the labels and the table to tick (✓) the **three** correct statements below. [3]

- Television 1 uses less energy per second than television 2
- The largest televisions always use the most energy
- The purchase cost of television 2 is 1.5 times that of television 3
- More expensive televisions always use less energy
- Television 3 uses 40 units more per year than television 4
- Televisions with the same energy rating, e.g A+, don't always have the same power
- ☐
- ☐
- ☐
- ☐
- ☐
- ☐



- (b) It is claimed that power is proportional to screen size. Use the data for television **1** and television **2** to determine if this claim is true. [3]
Space for calculations.

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- (c) (i) Use the equation:

$$\text{time (h)} = \frac{\text{units used (kWh)}}{\text{power (kW)}}$$

and data from the energy label to calculate how many hours the label suggests that television **2** is used for in 1 year. [2]

Hours used =

- (ii) Use an equation from page 2 to calculate the cost of using television **2** for 1 year if 1 unit (kWh) of electricity costs 16 p. Give your answer in pounds (£). [2]

Cost = £



Examiner
only

(iii) The expected lifespan of a television is 10 years. Simon concludes that it will be more cost effective to buy and run television **2** for 10 years but Sarah disagrees and claims that television **4** will be cheaper. Use the data to determine who is right. [3]
Space for calculations.

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(iv) Other than to save money, why should consumers be encouraged to choose appliances that use less energy? [2]

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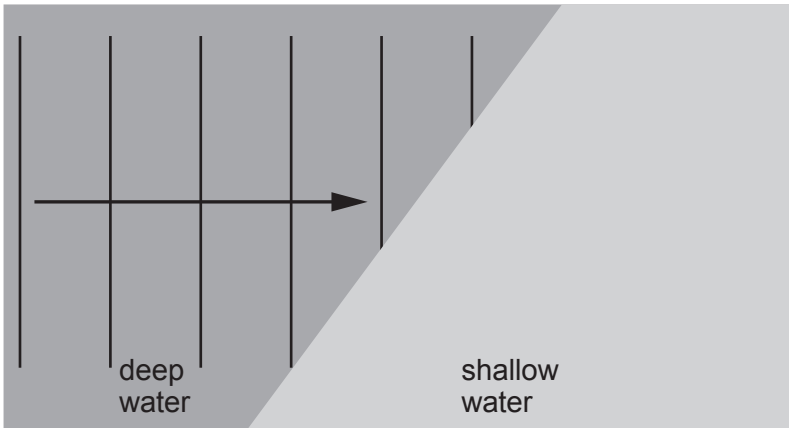
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3430UC01
07



2. A physics teacher demonstrates the refraction of water waves in a ripple tank.

(a) **Complete the diagram below** to show the wavefronts in the shallow water. [2]



(b) The wave generator in the deep water produces 5 waves in 10 s. Use an equation from page 2 and a measurement from the diagram to determine the speed of the waves in the deep water. Give your answer in m/s. [3]

Wave speed = m/s.

(c) State how the frequency, wavelength and wave speed will compare in deep and shallow water. [3]

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Examiner
only

3. Describe how the density of a small pebble, which is not a regular shape, can be determined. [6 QER]

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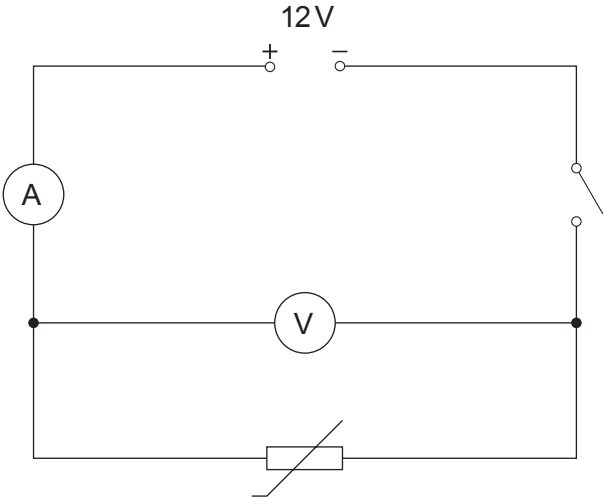
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09



4. A thermistor is a type of resistor whose resistance depends on temperature. Thermistors are often used as temperature sensors in applications where we want to monitor or control temperature. A group of students immerse a thermistor and a thermometer in water in order to determine the resistance at different temperatures. The circuit they use is shown below.



Their results are given below.

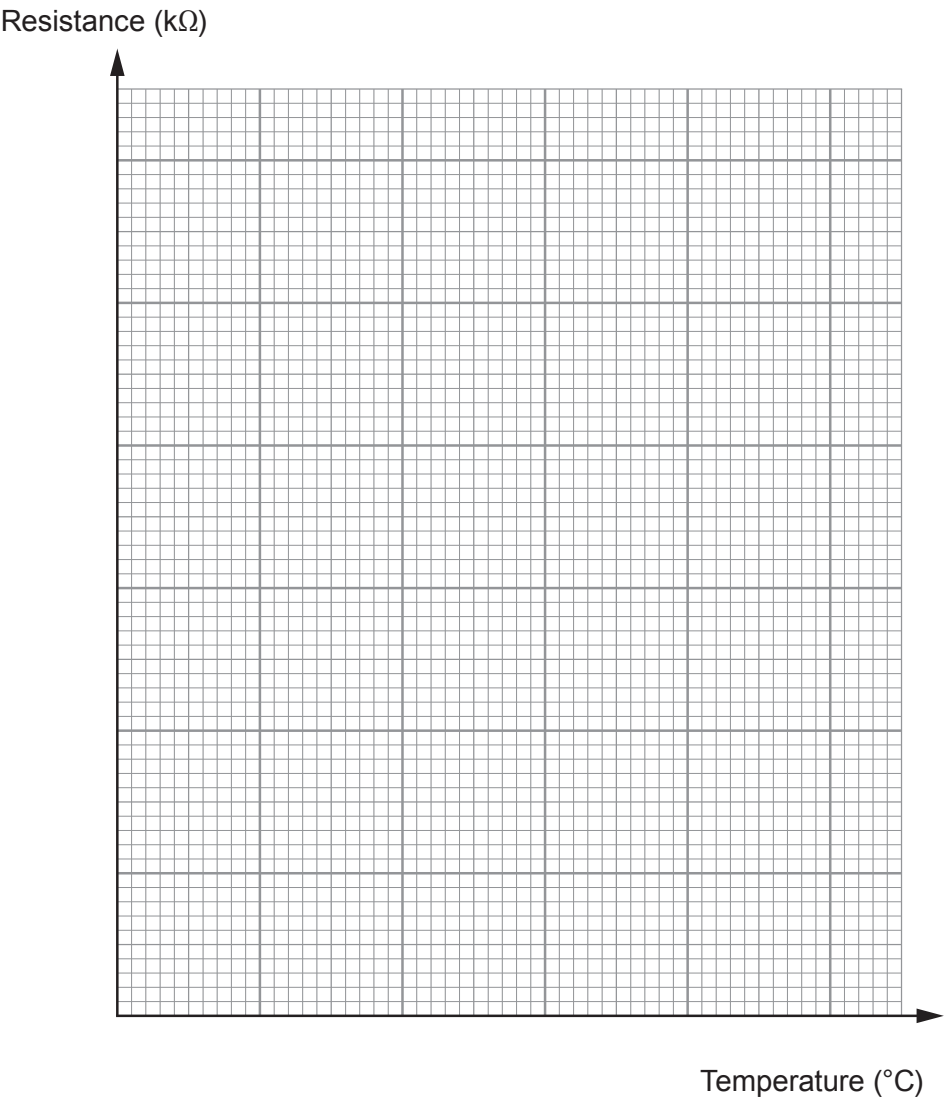
Temperature (°C)	Current (mA)	Voltage (V)	Resistance (kΩ)
20	0.97	12.0	12.4
40	2.22	12.0	5.41
60	4.80	12.0	2.50
80	9.23	12.0	1.30
100	17.14	12.0	0.70



Examiner
only

(a) (i) Plot the data on the grid below and draw a suitable line.

[4]



(ii) Describe the relationship between temperature and resistance.

[2]

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Examiner
only

- (b) (i) A resistor of resistance $5\text{ k}\Omega$ is now added to the original circuit, in parallel with the thermistor which is at a temperature of 30°C . The same 12 V power supply is used. Use your graph and equations from page 2 to calculate the total current in the circuit. [5]

Current = A

- (ii) Kim suggests that the current through both the resistor and the thermistor will decrease as the thermistor cools down. Explain whether or not she is correct. [3]

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14



5. A combined heat and power station uses the wasted heat energy from a power station to provide heating for local homes and businesses. This increases the efficiency of the energy transfers.

(a) Describe how electricity is produced in a fossil fuel power station. [3]

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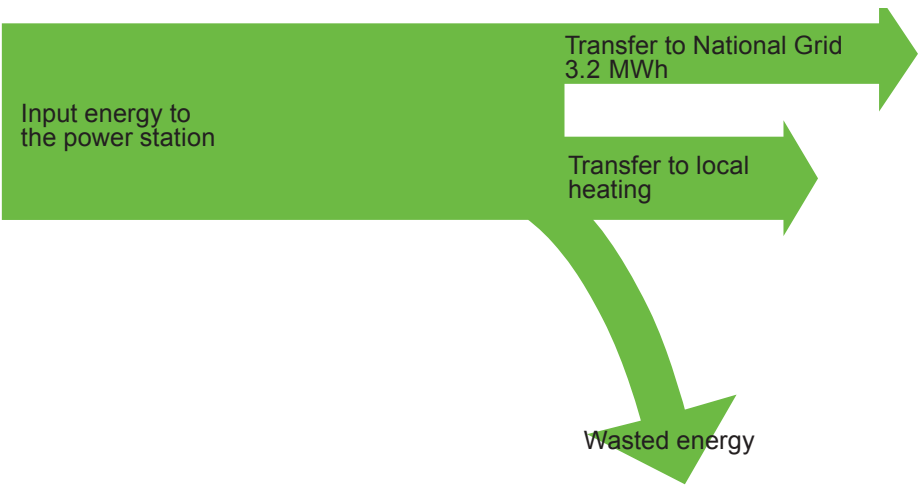
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(b) The Sankey diagram shows the energy transfers for a combined heat and power station in a housing development. 75 % of the input energy to the power station is transferred as useful energy, some to the National Grid and some to local heating. 40 % of the useful energy produced by the power station is transferred to the National Grid. Helen estimates the input energy to the power station to be 20 MWh. Determine if this estimate is correct. [3]



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Examiner
only

- (c) Another power station produces 1 200 MW at a voltage of 50 kV. This voltage is increased to 400 kV by a step-up transformer before being transmitted by the National Grid. Use an equation from page 2 to calculate the current in the wires of the grid. Assume that the transformer is 100% efficient. [3]

Current = A

9



6. Satellites in orbit around the Earth are used for a variety of purposes. One use is communications between base stations and satellites. Satellites for communications are usually put into geostationary orbits, orbiting above the equator. Satellites for other uses such as weather forecasting can be put in polar orbits. The speed of the satellite and therefore its orbital period, the time for one orbit of the Earth, is determined by its orbit height.

(a) Another kind of orbit is a geosynchronous orbit. State **one** similarity and **one** difference between a geostationary and a geosynchronous orbit. [2]

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(b) The table below gives data on the orbits of different satellites.

Satellite	Height above Earth (km)	Period (hours)
1	21 450	12.9
2	24 000	15
3	36 000	24
4	507	1.6
5	1 090	1.8
6	20 200	12

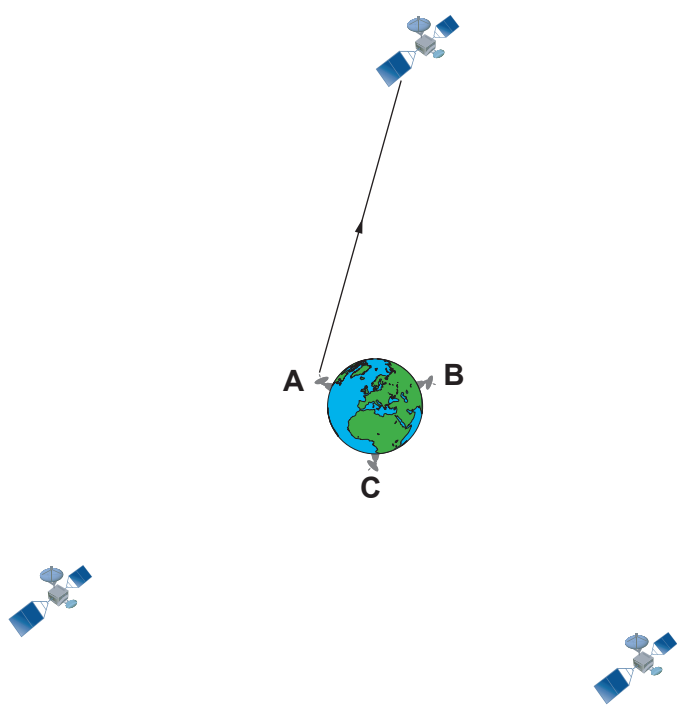
(i) State which satellite is in geostationary orbit. [1]

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Examiner
only

- (ii) A signal is sent from base station **A** to a geostationary satellite and is received at another base station after a time delay of 0.48 s. An incomplete diagram of the signal's path is shown.



Use an equation from page 2 to calculate the distance the signal has travelled. The speed of light, $c = 3 \times 10^8$ m/s. [3]

Distance = m

- (iii) Use your answer to (ii) and the information in the table to determine which base station, **A**, **B** or **C**, finally receives the signal. Show all your workings. [2]

Receiving base station =

END OF PAPER

8



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3430UC0-1



MONDAY, 20 JUNE 2022 – MORNING

SCIENCE (Double Award)

**Unit 3 – PHYSICS 1
HIGHER TIER**

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	15	
2.	13	
3.	6	
4.	7	
5.	6	
6.	13	
Total	60	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **5**.



JUN223430UC0101

Equations

current = $\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
total resistance in a series circuit	$R = R_1 + R_2$
total resistance in a parallel circuit	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$
energy transferred = power $\times$ time	$E = Pt$
power = voltage $\times$ current	$P = VI$
power = current ² $\times$ resistance	$P = I^2R$
% efficiency = $\frac{\text{energy (or power) usefully transferred}}{\text{total energy (or power) supplied}} \times 100$	
density = $\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
units used (kWh) = power (kW) $\times$ time (h) cost = units used $\times$ cost per unit	
wave speed = wavelength $\times$ frequency	$v = \lambda f$
speed = $\frac{\text{distance}}{\text{time}}$	

SI multipliers

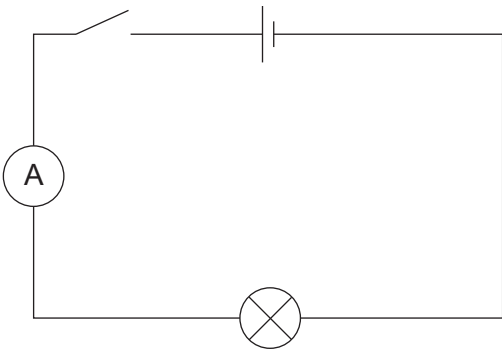
Prefix	Multiplier
p	1×10^{-12}
n	1×10^{-9}
μ	1×10^{-6}
m	1×10^{-3}

Prefix	Multiplier
k	1×10^3
M	1×10^6
G	1×10^9
T	1×10^{12}

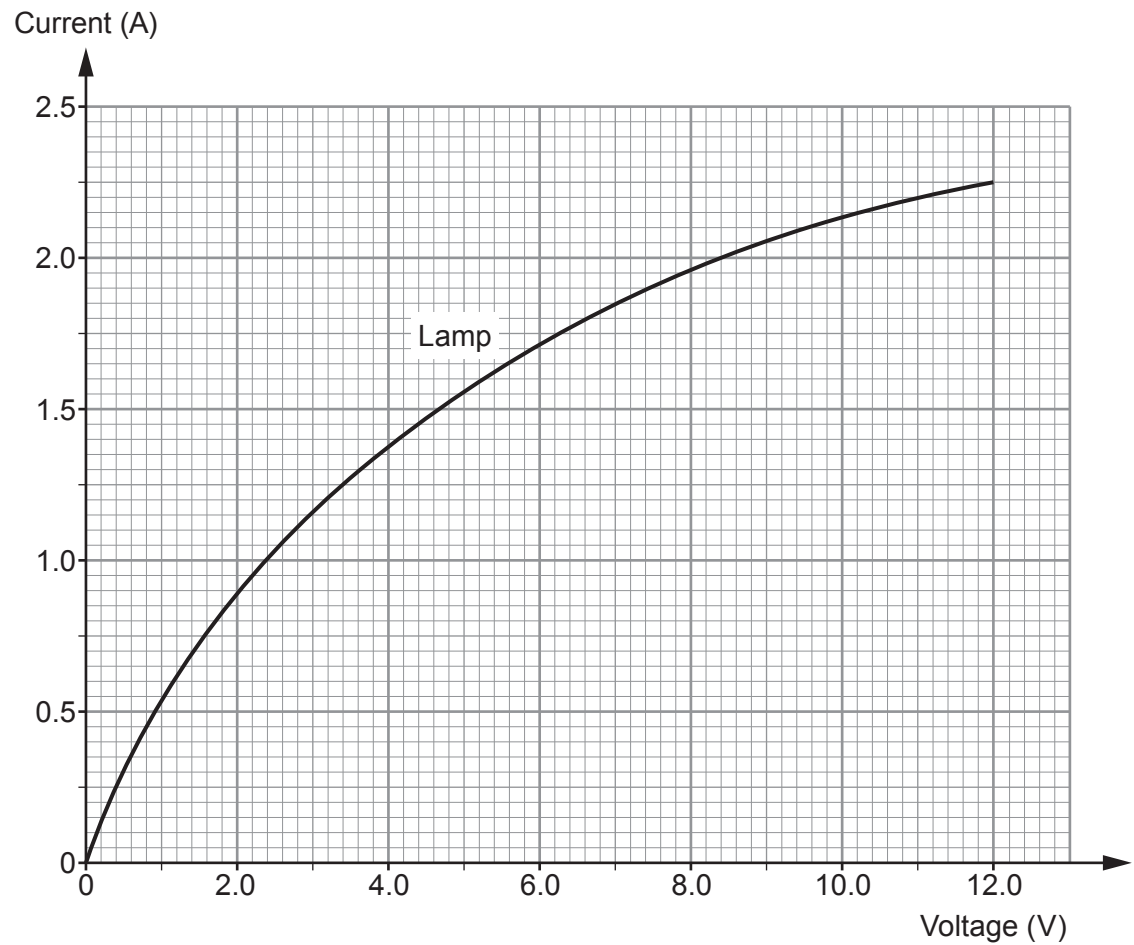


Answer **all** questions.

1. A group of pupils investigate the current-voltage (I - V) characteristics of different components.
- (a) The first component they investigate is a lamp.
Part of the circuit used is shown below.
Add a variable resistor **and** voltmeter to the circuit diagram. [2]



- (b) They draw a graph from their results for the **lamp**. It is shown below.



- (i) One student suggests that as the current through the lamp **doubles** the voltage **triples**.
Use pairs of data within the range 0.5 A to 2.0 A from the graph to explain whether you agree with the student. [3]

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- (ii) Use the equation:
- power = voltage × current
- and information from the graph to calculate the **maximum** power produced by the lamp. [3]

Power = W

- (c) The experiment is repeated with a 6 Ω resistor but the results are lost.

- (i) Use the equation:

$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

to calculate the current through the 6 Ω resistor at 12 V. [2]

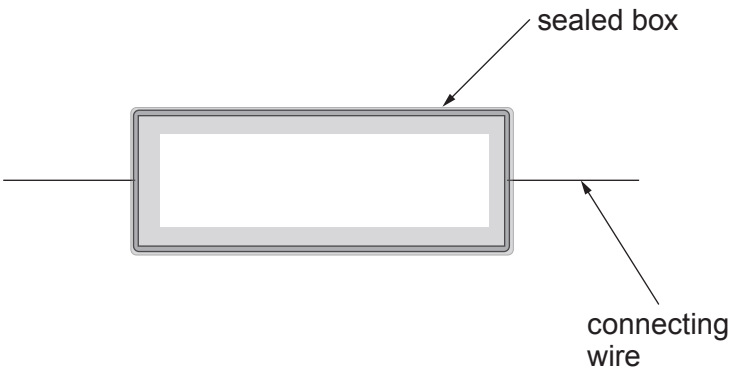
Current = A

- (ii) **Draw the line for this resistor** on the grid on the previous page. [1]

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05



(d) The students are given a sealed box containing another component.



They are asked to confirm whether the hidden component is a diode.

(i) Describe how they would use the circuit shown in part (a) to confirm whether or not it is a diode. [3]

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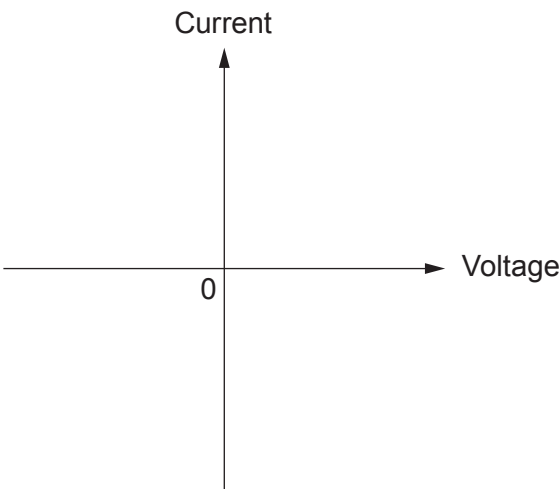
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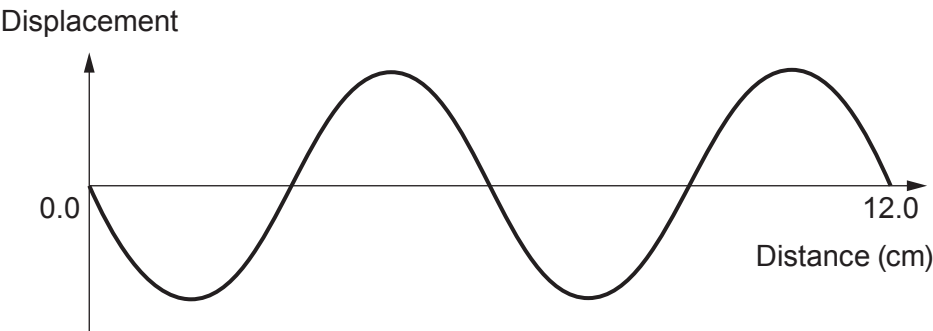
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(ii) **Sketch** the I - V graph that you would expect to obtain for a diode on the grid below. [1]



2. Microwaves travel through space at a speed of $3 \times 10^8 \text{ m/s}$ and are one part of the electromagnetic spectrum.

(a) The diagram represents microwaves produced at a certain frequency.



(i) Use the information in the diagram to calculate the wavelength of these microwaves. [1]

Wavelength = cm

(ii) Use your answer above and the equation:

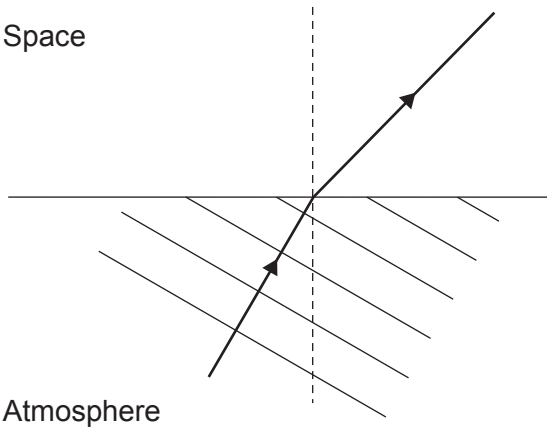
wave speed = wavelength $\times$ frequency

to calculate the frequency of these microwaves. [3]

Frequency = Hz



(b) Microwaves are refracted as they travel from the atmosphere into space.



Complete the diagram to show the refracted wavefronts in space. [3]

(c) Geostationary artificial satellites are used to send microwave signals around the world.

(i) Explain why a satellite in a geostationary orbit above the equator appears to stay in a fixed place above the surface of the Earth even though both the satellite and the Earth are constantly moving. [2]

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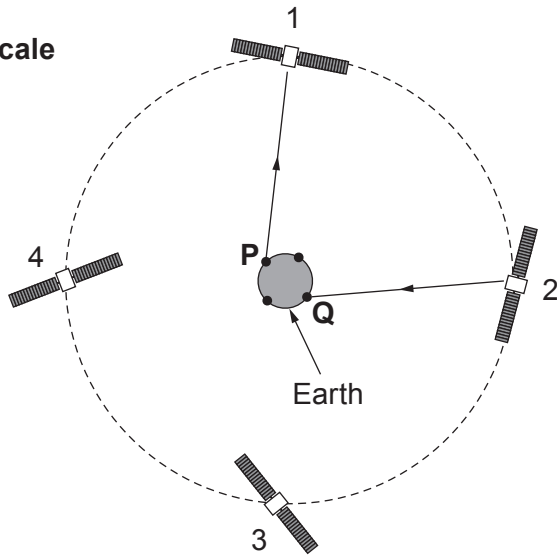
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09



- (ii) The diagram shows 4 satellites in geostationary orbit around the Earth. A signal is to be sent from Earth station **P** to **Q** using the satellites 1 and 2. The first and last parts of the path are shown.

Diagram not to scale



Use an equation from page 2 to calculate the time taken for the signal to travel from **P** to **Q**.

The height of a geostationary satellite above the Earth is $3.6 \times 10^7 \text{ m}$. [3]
(Speed of light, $c = 3 \times 10^8 \text{ m/s}$)

Time = s

- (iii) It was suggested that if the signal had been sent in the opposite direction from **Q** to **P** via satellites **3** and **4** then it would have arrived quicker. Explain whether you agree with this suggestion. [1]

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Examiner
only

3. A ring main is used to connect sockets to the fuse box.
The cables include live, neutral and Earth wires.
The fuse box includes miniature circuit breakers (mcb) and residual current circuit breakers (rccb).

(a) Describe the function of each of the following wires. [3]

live:

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neutral:

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Earth:

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(b) Explain the differences between an mcb and an rccb. [3]

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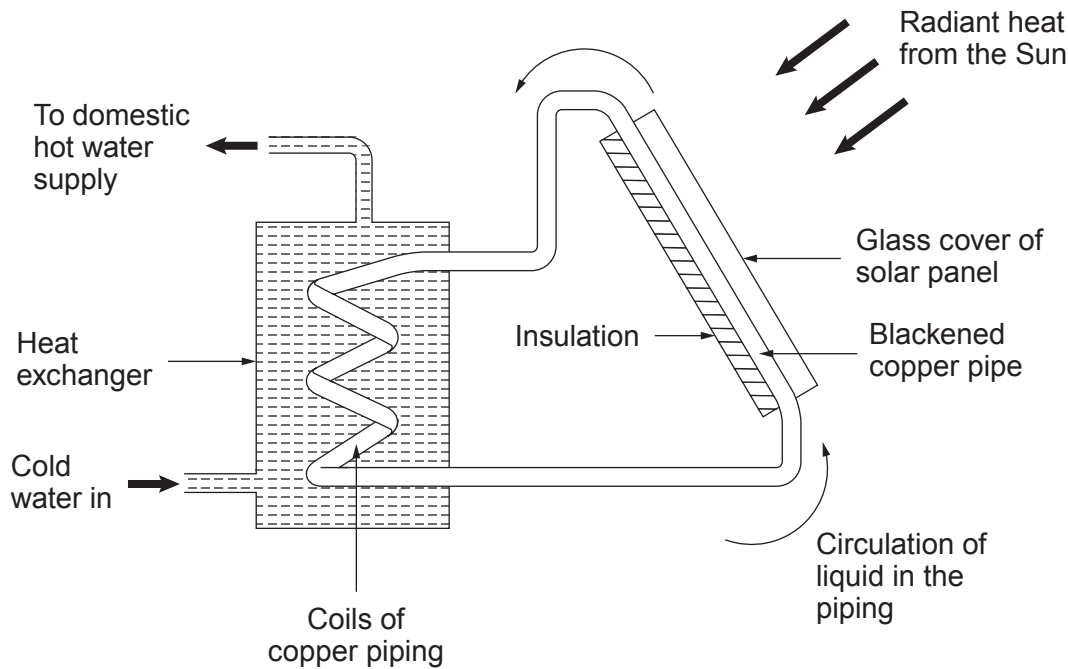
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4. The diagram shows a solar heating panel that is installed on a roof. The liquid in the solar panel becomes heated by radiation from the Sun. The heated liquid passes through coils of copper piping inside a heat exchanger resulting in water heating up.



- (a) Explain, **in terms of particles**, how heat energy conducts through the walls of the copper piping in the heat exchanger. [3]

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- (b) Explain why the liquid in the solar panel circulates in the direction shown. [2]

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Examiner
only

- (c) In good sunlight, each square metre of the solar panel receives 960 J of energy per second from the Sun. Only $\frac{1}{3}$ of this energy is used to heat the liquid.

Calculate the amount of energy given to the liquid per second if the panel has an area of 2 m². [2]

Energy given to the liquid per second = J

7



Examiner
only

5. Describe the processes involved in generating electricity in a gas fired power station **and** explain how this electricity is supplied to consumers efficiently and safely through the National Grid. [6 QER]

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6. A dairy farm was using 15 000 kWh of electricity every month from the National Grid. The cost of electricity to the farm was 20 p per unit. The farmer decided to spend £152 400 on a biogas generator which uses cow dung to generate electricity. When working at maximum output, it was expected to reduce his demand on the National Grid to only 3 000 kWh a month.

(a) (i) Use an equation from page 2 to calculate the expected monthly savings in £. [3]

Savings = £

(ii) Calculate the payback time for the biogas generator. [1]

Payback time = months

(b) The biogas generator needs to produce 144 000 kWh of electricity per year. The farmer owns 150 cows. Each cow produces 200 kg of dung per week, but the farmer is only able to collect 60 kg of this. Each 1 kg of dung produces 0.05 m³ of methane gas. Each 1 m³ of methane gas input to the generator has an energy value of 5.3 kWh. Each 1 m³ of methane gas when burned produces an output of 1.9 kWh of electricity.

(i) Use the information above and an equation from page 2 to calculate the % efficiency of using methane gas to produce electricity. [2]

% efficiency =



Examiner
only

- (ii) The farmer thinks that there will be enough cow dung to produce the required amount of electricity (144 000 kWh).
Use the information opposite to explain whether the farmer is correct.
Show all your workings in the space below. (1 year = 52 weeks) [5]

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- (c) When cow dung decomposes it produces methane gas.
When methane gas burns, it produces the greenhouse gases carbon dioxide and water vapour.
An online article contains this information about methane:

“While carbon dioxide is said to be the major contributor to the greenhouse effect, methane is roughly 30 times more effective as a heat-trapping gas.”

Explain whether collecting cow dung to use in a biogas generator benefits efforts to reduce human impact on the greenhouse effect. [2]

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END OF PAPER

13



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3430UC0-1



MONDAY, 19 JUNE 2023 – AFTERNOON

SCIENCE (Double Award)
Unit 3 – PHYSICS 1

HIGHER TIER

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	6	
3.	3	
4.	9	
5.	8	
6.	6	
7.	12	
8.	7	
Total	60	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question 6.



JUN233430UC0101

Equations

current = $\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
total resistance in a series circuit	$R = R_1 + R_2$
total resistance in a parallel circuit	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$
energy transferred = power $\times$ time	$E = Pt$
power = voltage $\times$ current	$P = VI$
power = current ² $\times$ resistance	$P = I^2R$
% efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$	
density = $\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
units used (kWh) = power (kW) $\times$ time (h) cost = units used $\times$ cost per unit	
wave speed = wavelength $\times$ frequency	$v = \lambda f$
speed = $\frac{\text{distance}}{\text{time}}$	

SI multipliers

Prefix	Symbol	Conversion factor	Multiplier
pico	p	divide by 1 000 000 000 000	1×10^{-12}
nano	n	divide by 1 000 000 000	1×10^{-9}
micro	μ	divide by 1 000 000	1×10^{-6}
milli	m	divide by 1000	1×10^{-3}
centi	c	divide by 100	1×10^{-2}

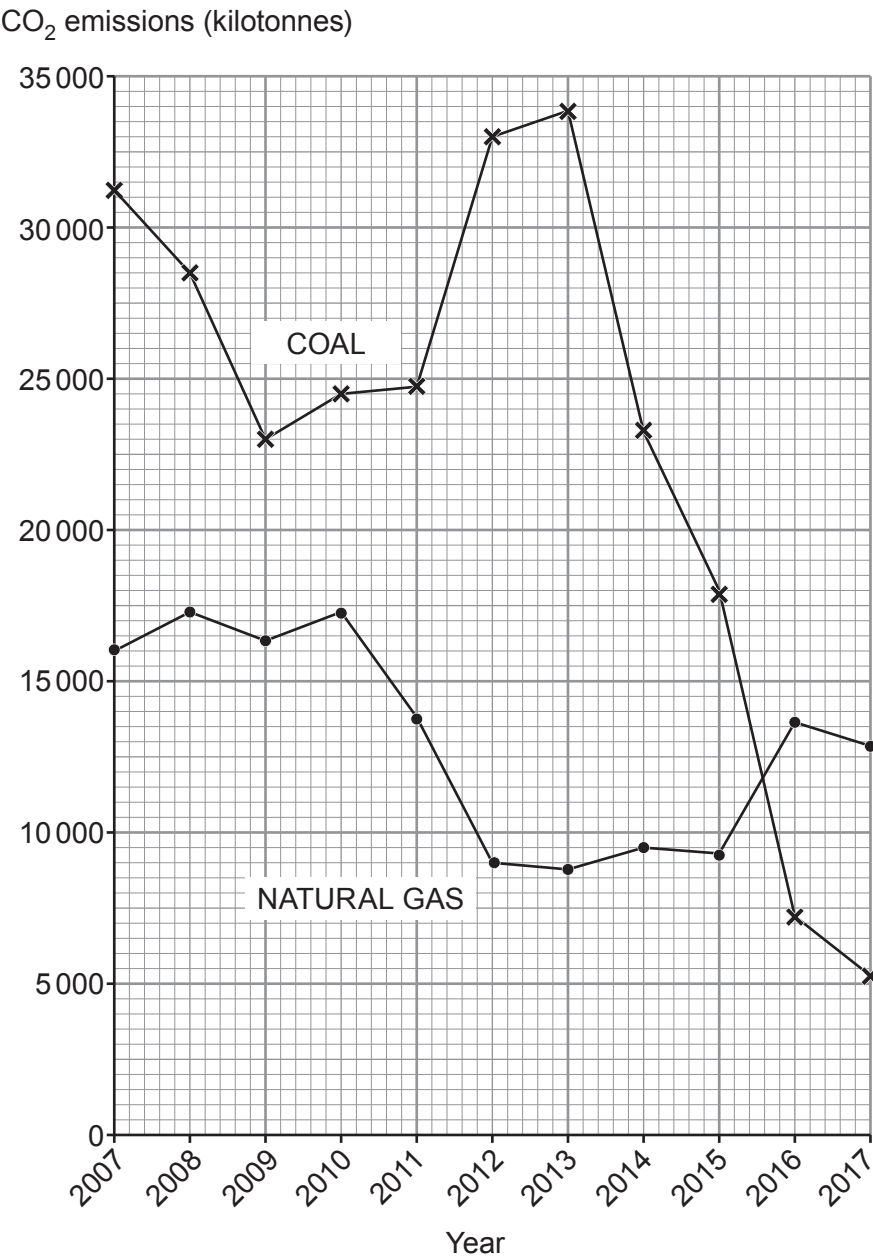
kilo	k	multiply by 1000	1×10^3
mega	M	multiply by 1 000 000	1×10^6
giga	G	multiply by 1 000 000 000	1×10^9
terra	T	multiply by 1 000 000 000 000	1×10^{12}



Answer **all** questions.

1. The UK government wants to reduce the CO₂ emissions from power stations.

The graphs below show the CO₂ emissions from coal and natural gas power stations between the years 2007 and 2017.



- (a) Look at the data for **2012**. Calculate the difference in the emissions of CO₂ from coal and natural gas. [2]

difference = kilotonnes of CO₂

- (b) Between the years 2014 and 2015, the emission of CO₂ from coal fell by 5500 kilotonnes.
- State between which other years the emission of CO₂ from coal fell **at the same rate**. [1]
- Years and

- (c) State **two** benefits of reducing CO₂ in the atmosphere. [2]

1.
2.

- (d) (i) Nuclear power stations provide up to 20% of the present UK demand for electricity.
- Gas provides up to 50%.
- One student, Seren, says that a graph for the CO₂ emissions from nuclear power stations would be the same shape as for gas but always lower.
- Explain whether you agree with Seren. [2]

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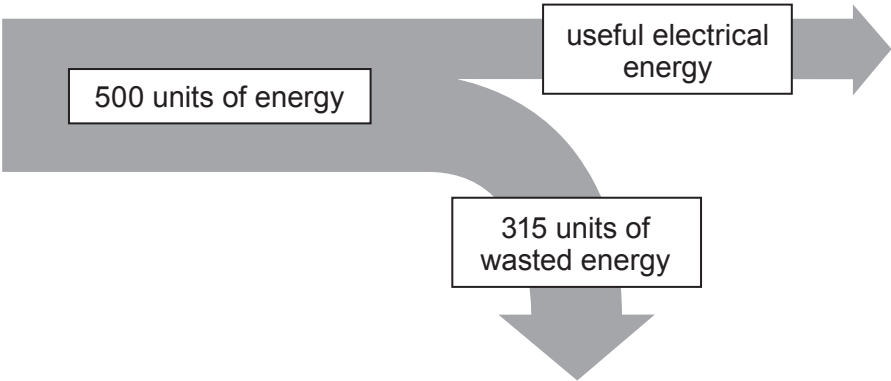
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05



(ii) The Sankey diagram shows the energy input and output for a power station.



Seren looks at the diagram and calculates that the power station is 63% efficient.

Explain whether you agree with Seren. [2]

Space for calculation.

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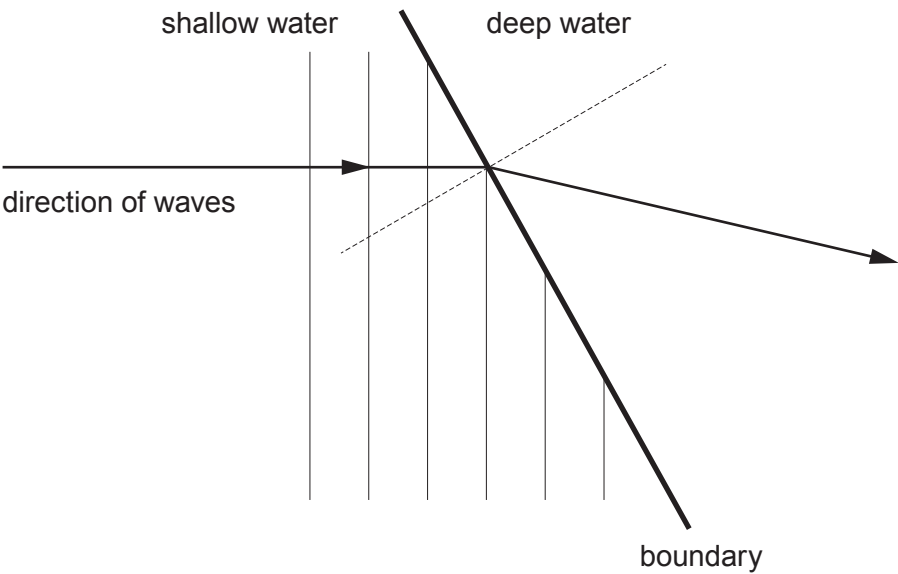
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2. Students study refraction of waves in a ripple tank.

They set up the tank to show waves going from shallow water into deeper water.

The diagram shows a number of wavefronts approaching a boundary between shallow and deep water.

The first four wavefronts have reached the boundary.

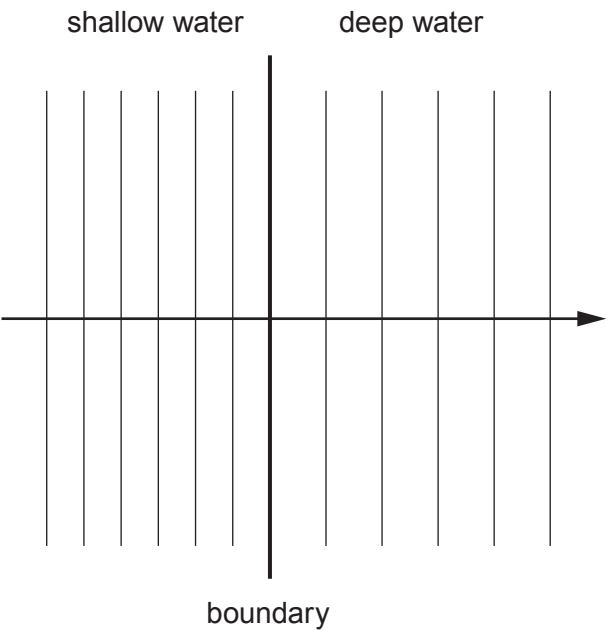


(a) **Complete the diagram** to show the wavefronts in the deep water.

[3]



(b) In a separate experiment, the wavefronts pass over the boundary in the way shown below.



Sam says that the speed of the waves is greater in deep water than shallow water.

Explain whether you agree with his statement.

[3]

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09



3. A washing machine uses 0.54 kWh per wash cycle.

(a) A customer uses it for 200 wash cycles per year.

Use the equation:

$$\text{cost} = \text{units used (kWh)} \times \text{cost per unit}$$

to calculate the cost of using the washing machine **for 1 year in £.** [2]

The cost of a unit is £0.30.

cost = £

(b) The wash cycle lasts for a time of 4 hours.

Use the information above and the equation:

$$\text{mean power (kW)} = \frac{\text{units used (kWh)}}{\text{time (h)}}$$

to calculate the mean power of the washing machine during one wash cycle. [1]

mean power = kW

3

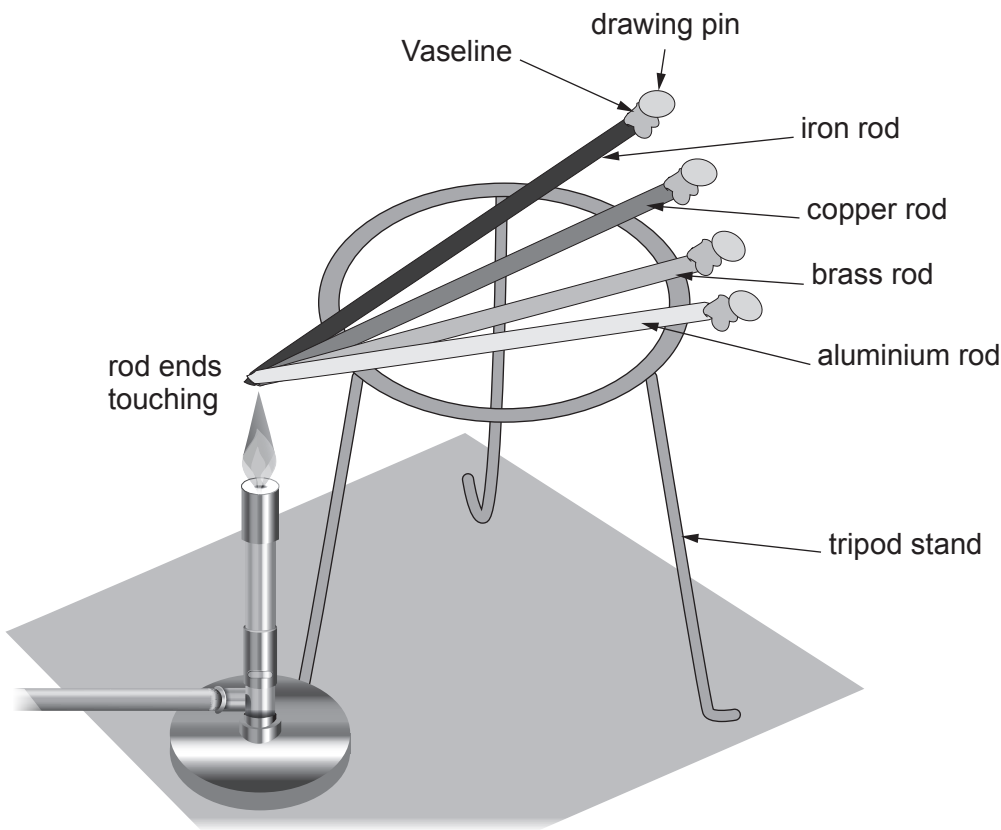


4. (a) (i) The apparatus shown can be used to compare how well different metals conduct heat.

Explain:

- how the apparatus is used
- how the results are used to show that copper is the best conductor.

[3]



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(ii) Describe, in terms of the behaviour of **particles**, how heat energy is conducted through the metal rods.

[2]

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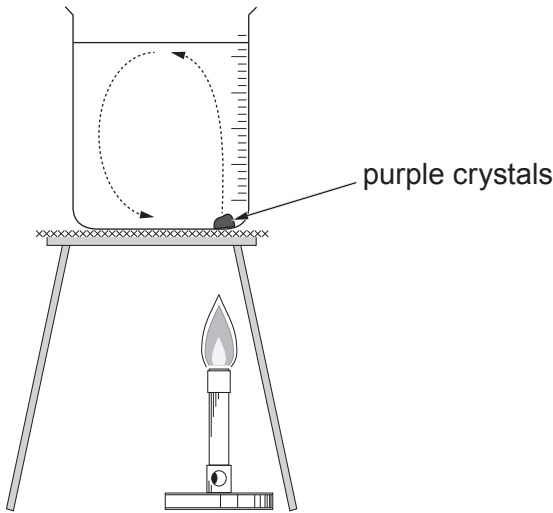
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(b) A teacher demonstrates the process of convection in liquids.

Some purple crystals are placed in the bottom corner of a beaker that is filled with water.

The water is then heated and a current of coloured water is seen to move as shown in the diagram below.



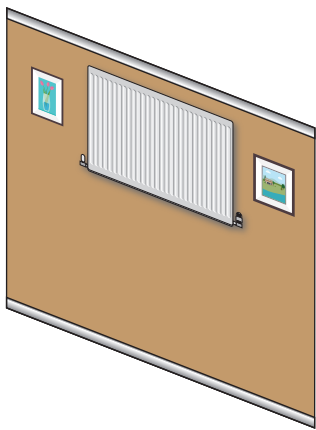
(i) Explain why the coloured water moves in the way shown above. [2]

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(ii) Explain why it is not suitable to put a hot radiator at the top of a wall to warm the room. [2]



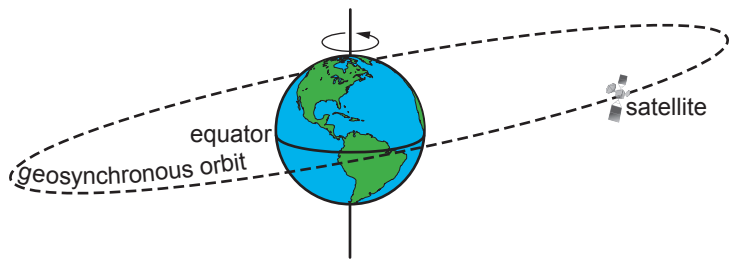
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5. The diagram shows an example of a satellite in geosynchronous orbit around the Earth.

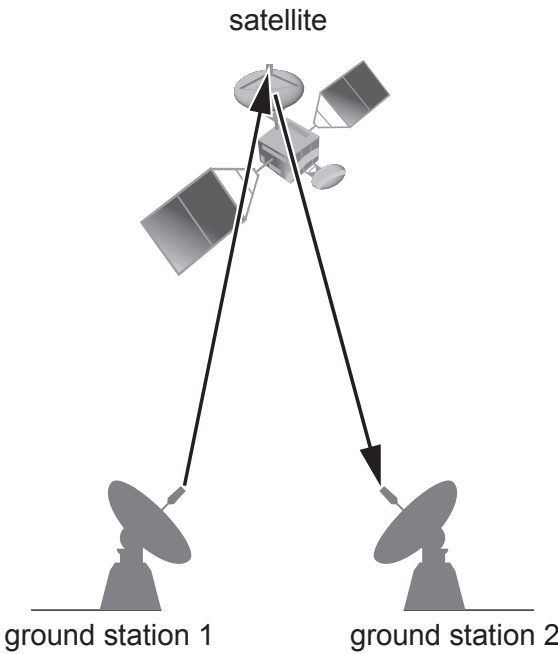


(a) State **one** way in which a geostationary orbit is different to the one shown above. [1]

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(b) The diagram shows a signal sent from one ground station to a satellite which is then received back at a second ground station.



Examiner
only

(i) Use the equation:

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

to calculate the time interval between a microwave signal being sent from Earth to one of the satellites 36 000 000 m away and it being received back again. [3]
(speed of light, $c = 3 \times 10^8$ m/s)

time = s

(ii) Microwaves travel at the speed of light and have a range of wavelengths between 0.002 m and 1 m.
Use an equation from page 2 to calculate the **maximum** frequency of microwave signals. [3]

maximum frequency = Hz

(iii) Name **one** type of electromagnetic radiation which has frequencies smaller than those of microwaves. [1]

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8



Examiner
only

6. Describe how you would find the density of a small irregular-shaped rock.
You may assume that all the usual laboratory apparatus is available for you to use. [6 QER]

Include in your answer:

- the apparatus you would use
- a description of the method you would use to obtain results
- how the density would be calculated.

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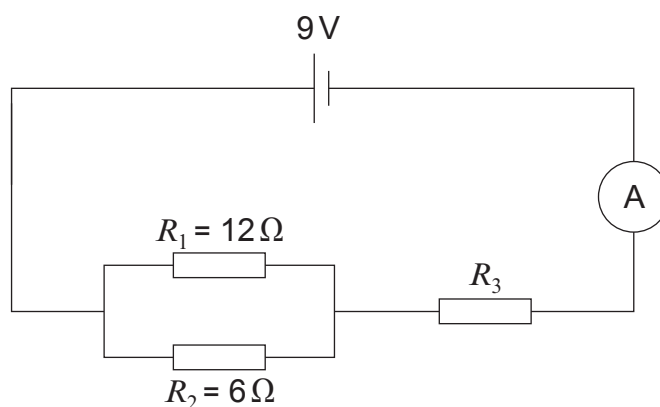
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6



7. A group of students set up the following circuit.



- (a) (i) Use the equation:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

to calculate the total resistance of resistors R_1 and R_2 in parallel.

[3]

total resistance of R_1 and R_2 = Ω

- (ii) The total resistance of the circuit is 6Ω .

Use an equation from page 2 to calculate the resistance of resistor R_3 .

[1]

resistance of R_3 = Ω

- (iii) Use an equation from page 2 to calculate the current through the ammeter.

[2]

current = A



Examiner
only

(iv) Calculate the voltage across the resistor R_3 . [3]

voltage across R_3 = V

(b) One of the students, Katrina, connects a different circuit, without R_1 and R_2 .
The circuit only contains R_3 in series with the battery.
She correctly calculates that the current from the battery would be 4.5 A.
She **claims** that the new circuit would transfer 45 J of energy in 10 s.
By using equations from page 2, explain whether her claim is correct. [3]

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12



8. Germany is building the world’s first system in which wind turbines are combined with a hydroelectric pumped storage system.

(a) State the main disadvantage of relying on energy from wind turbines. [1]

(b) When the output from the wind farm is not required by the German National Grid, the energy from the wind is used to pump water from a low level to a higher level.

The water at the higher level can then be used to generate electricity when required, just as in a conventional hydroelectric pumped storage system.



When generating, the power output from the hydroelectric pumped storage system is 16 MW.

The mean power used by homes in Germany was 0.43 kW in 2017 (which was the lowest on record up to that time).

(i) Suggest a possible reason why the 0.43 kW figure is lower than for any previous year. [1]



Examiner
only

- (ii) Calculate the number of homes in Germany in 2017, that could be supplied using the power from the hydroelectric pumped storage system. [3]

number of homes =

- (c) The wind turbines have a maximum power output of 13.6 MW.
- They provide energy to the water when slowly pumping it to a higher level.
- When needed, the hydroelectric pumped storage system can quickly generate electricity.
- Scientists state it is possible to get a greater **power** from the hydroelectric pumped storage system (16 MW) than the power supplied to it by the wind turbines.

By considering the equation:

$$\text{power} = \frac{\text{energy transferred}}{\text{time}}$$

explain why the claim is true. [2]

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END OF PAPER

7

